

OICPC455

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Machine Learning Lab

Quality Laboratory Manual

Experiment No. 2

Study and Implementation of Multiple Linear Regression



Course Instructor -
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Experiment No. 2

Title of Experiment: To study and implement Multiple Linear Regression

Aim of Experiment: To implement and understand the working of Simple Multiple Regression, basic predictive modeling technique in Machine Learning

System Requirements – Win 8 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Needed for Experiment – Jupyter Notebook/ Anaconda Navigator/ Google Colaboratory/ Spyder, Python 3.x [With libraries such as Numpy, Pandas, Matplotlib and Scikit-Learn]

Experiment Outcomes –

1. Understand the principles and applications of simple linear regression
2. Differentiate between simple and multiple linear regression
3. Analyze and implement the difference between univariate and multivariate analysis
4. To implement linear regression in Python

Theory –

Multiple Linear Regression is an extension of Simple Linear regression as it takes more than one predictor variable to predict the response variable.

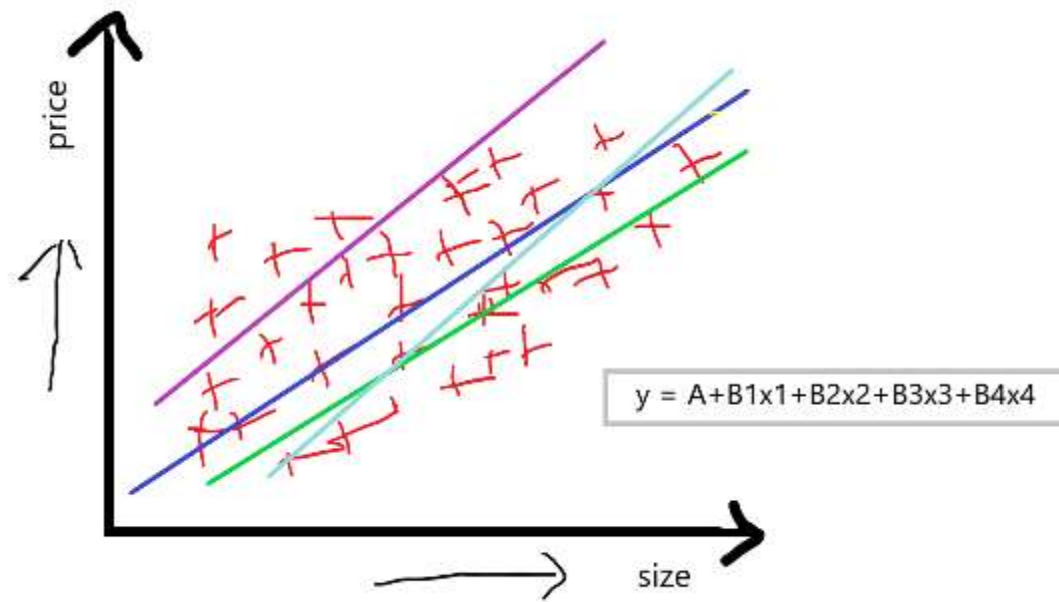
Multiple Linear Regression (MLR) is basically indicating that we will have many features such as f1, f2, f3, f4, and our output feature f5.

f1 is the size of the house,
f2 is bad rooms in the house,
f3 is the locality of the house,
f4 is the condition of the house, and
f5 is our output feature, which is the price of the house

The multiple independent features also make a huge impact on the price of the house, meaning the price can vary from feature to feature. When we are discussing multiple linear regression, then the equation of simple linear regression $y = a + bx$ is converted to something like:

$$y = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

If we have one dependent feature and multiple independent features then basically call it a multiple linear regression.



Our aim in using the multiple linear regression is that we have to compute A, which is an intercept. The key parameters B1, B2, B3, and B4 are the slopes or coefficients concerning this independent feature. This basically indicates that if we increase the value of x_1 by 1 unit, then B1 will tell you how much it will affect the price of the house. The others B2, B3, and B4, also work similarly.

The general form of the multiple linear regression equation is:

$$y = \beta_0 + \beta_1 \times x_1 + \beta_2 \times x_2 + \dots + \beta_n \times x_n$$

Where,

y is the dependent variable.

$x_1, x_2, \dots, x_n$ are the independent variables.

β_0 is the intercept.

$\beta_1, \beta_2, \dots, \beta_n$ are the coefficients representing the impact of each independent variable on y

The objective is to determine the coefficients ($\beta_0, \beta_1, \dots, \beta_n$) that minimize the sum of squared errors (SSE) between the observed and predicted values.

Procedure to implement Multiple Linear Regression:

1. **Data Collection:** Obtain a dataset with one dependent variable and multiple independent variables. For this experiment, you can use a dataset that includes factors such as years of experience, education level, and age to predict salary.
2. **Data Preprocessing:**
 - a. Load the dataset into a Pandas DataFrame.
 - b. Handle missing values, outliers, and encode categorical variables if necessary.
 - c. Normalize or standardize the data if needed.

3. **Exploratory Data Analysis (EDA):** Visualize relationships between the dependent and independent variables using scatter plots and correlation matrices.
4. **Splitting the Data:** Split the dataset into training and test sets to evaluate the model's performance.
5. **Implementing Multiple Linear Regression:**
 - a. Use the Scikit-learn library to fit a multiple linear regression model.
 - b. Train the model using the training data.
6. **Model Prediction:**
 - a. Use the trained model to make predictions on the test data.
 - b. Compare predicted values with actual values to assess accuracy.
7. **Model Evaluation:** Evaluate the model using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2)

Sample Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Step 1: Data Collection (Example dataset)**# Dataset: Factors influencing Salary (Experience, Education Level, Age)**

```
data = {
    'Experience': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Education_Level': [1, 1, 2, 2, 3, 3, 3, 4, 4, 4],
    'Age': [22, 25, 28, 30, 32, 35, 38, 40, 42, 45],
    'Salary': [30, 35, 40, 45, 50, 55, 60, 65, 70, 75]
}
```

```
df = pd.DataFrame(data)
```

Step 2: Data Preprocessing**# No missing values or categorical variables in this simple dataset.****# Step 3: Exploratory Data Analysis (EDA)****# Plotting scatter plots for each independent variable against the dependent variable (Salary)**

```
plt.figure(figsize=(15, 5))
```

```
plt.subplot(1, 3, 1)
```

```
plt.scatter(df['Experience'], df['Salary'], color='blue')
plt.xlabel('Experience (years)')
plt.ylabel('Salary (k$)')
plt.title('Experience vs Salary')

plt.subplot(1, 3, 2)
plt.scatter(df['Education_Level'], df['Salary'], color='green')
plt.xlabel('Education Level')
plt.ylabel('Salary (k$)')
plt.title('Education Level vs Salary')

plt.subplot(1, 3, 3)
plt.scatter(df['Age'], df['Salary'], color='red')
plt.xlabel('Age (years)')
plt.ylabel('Salary (k$)')
plt.title('Age vs Salary')

plt.show()

# Step 4: Splitting the Data
X = df[['Experience', 'Education_Level', 'Age']] # Independent
variables
y = df['Salary'] # Dependent variable

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=0)

# Step 5: Implementing Multiple Linear Regression
regressor = LinearRegression()
regressor.fit(X_train, y_train)

# Step 6: Model Prediction
y_pred = regressor.predict(X_test)

# Comparing Actual vs Predicted
comparison_df = pd.DataFrame({'Actual': y_test, 'Predicted':
y_pred})
print(comparison_df)

# Step 7: Model Evaluation
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse}")
print(f"Root Mean Squared Error: {rmse}")
```

```
print(f"R^2 Score: {r2}")
```

Coefficients of the model

```
print(f"Intercept ( $\beta_0$ ): {regressor.intercept}")  
print(f"Coefficients ( $\beta_1, \beta_2, \beta_3$ ): {regressor.coef}")
```

Observations -

- Record the predicted salaries for the test data
- Observe how well the model predicts the salary based on the independent variables

Conclusion –

Hence, the experiment provides students with hands-on experience in implementing and analyzing a Multiple Linear Regression model, a foundational technique in data science and Machine Learning

References –

a. Textbook –

- i. Machine Learning with Python – An approach to Applied ML – Abhishek Vijayvargiya, BPB Publications, 1st Edition 2018
- ii. Machine Learning, Tom Mitchell, McGraw Hill Education, 1st Edition 1997

b. Online references –

- i. <https://www.analyticsvidhya.com/blog/2021/05/multiple-linear-regression-using-python-and-scikit-learn/>
- ii. <https://www.javatpoint.com/multiple-linear-regression-in-machine-learning>
- iii. <https://towardsdatascience.com/multiple-linear-regression-for-manufacturing-analysis-c057d4af718b>

Expected Oral Questions –

1. What is multiple Linear Regression?
2. What are the key assumptions of multiple Linear Regression?
3. What is the purpose of including multiple independent variables in a regression model?
4. What is the purpose of using a regression line in Simple Linear Regression?
5. What diagnostic plots would you use to assess the fit of a multiple Linear Regression?
6. How do you validate the performance of a multiple Linear Regression?
7. What are the consequences of violating the assumptions of multiple Linear Regression?
8. What is the formula/equation for multiple Linear Regression?
9. What are the applications for multiple Linear Regression?

10. State a basic example for multiple Linear Regression?

FAQ's in Interview –

1. What distinguishes multiple linear regression from simple linear regression?
2. In a multiple linear regression model, if you have 'p' predictors, what is 'p'?
3. How do you interpret a multiple linear regression model?
4. What are the basic assumptions of multiple Linear Regression algorithm?
5. How do you interpret the coefficients of multiple Linear Regression?

